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DSC640: Weeks 1&2

Milestone 1

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df = pd.read_excel("nationaldatabaseofchildcareprices.xlsx")
df.head()
```

```
Out[2]:
```

	State_Name	State_Abbreviation	County_Name	County_FIPS_Code	StudyYear	UNR_16
0	Alabama	AL	Autauga County	1001	2008	5.42
1	Alabama	AL	Autauga County	1001	2009	5.93
2	Alabama	AL	Autauga County	1001	2010	6.21
3	Alabama	AL	Autauga County	1001	2011	7.55
4	Alabama	AL	Autauga County	1001	2012	8.60

5 rows × 227 columns



```
In [10]: print(list(df.columns))
```

```
[ 'State_Name', 'State_Abbreviation', 'County_Name', 'County_FIPS_Code', 'StudyYear',
'UNR_16', 'FUNR_16', 'MUNR_16', 'UNR_20to64', 'FUNR_20to64', 'MUNR_20to64', 'FLFPR_2
0to64', 'FLFPR_20to64_Under6', 'FLFPR_20to64_6to17', 'FLFPR_20to64_Under6_6to17', 'M
LFPR_20to64', 'PR_F', 'PR_P', 'MHI', 'ME', 'FME', 'MME', 'MHI_2018', 'ME_2018', 'FME
_2018', 'MME_2018', 'TotalPop', 'OneRace', 'OneRace_W', 'OneRace_B', 'OneRace_I', 'O
neRace_A', 'OneRace_H', 'OneRace_Other', 'TwoRaces', 'Hispanic', 'Households', 'H_Un
der6_BothWork', 'H_Under6_FWork', 'H_Under6_MWork', 'H_Under6_SingleM', 'H_6to17_Bot
hWork', 'H_6to17_Fwork', 'H_6to17_Mwork', 'H_6to17_SingleM', 'EMP_M', 'MEMP_M', 'FEM
P_M', 'EMP_Service', 'MEMP_Service', 'FEMP_Service', 'EMP_Sales', 'MEMP_Sales', 'FEM
P_Sales', 'EMP_N', 'MEMP_N', 'FEMP_N', 'EMP_P', 'MEMP_P', 'FEMP_P', 'iUNR_16', 'iFUN
R_16', 'iMUNR_16', 'iUNR_20to64', 'iFUNR_20to64', 'iMUNR_20to64', 'iFLFPR_20to64',
'iFLFPR_20to64_Under6', 'iFLFPR_20to64_6to17', 'iFLFPR_20to64_Under6_6to17', 'iMLFPR
_20to64', 'iPR_F', 'iPR_P', 'iMHI', 'iME', 'iFME', 'iMME', 'iMHI_2018', 'iME_2018',
'iFME_2018', 'iMME_2018', 'iTotalPop', 'iOneRace', 'iOneRace_W', 'iOneRace_B', 'iOne
Race_I', 'iOneRace_A', 'iOneRace_H', 'iOneRace_Other', 'iTwoRaces', 'iHispanic', 'iH
ouseholds', 'iH_Under6_BothWork', 'iH_Under6_FWork', 'iH_Under6_MWork', 'iH_Under6_S
ingleM', 'iH_6to17_BothWork', 'iH_6to17_Fwork', 'iH_6to17_Mwork', 'iH_6to17_Single
M', 'iEMP_M', 'iMEMP_M', 'iFEMP_M', 'iEMP_Service', 'iMEMP_Service', 'iFEMP_Servic
e', 'iEMP_Sales', 'iMEMP_Sales', 'iFEMP_Sales', 'iEMP_N', 'iMEMP_N', 'iFEMP_N', 'iEM
P_P', 'iMEMP_P', 'iFEMP_P', 'MCBto5', 'MC6to11', 'MC12to17', 'MC18to23', 'MC24to29',
'MC30to35', 'MC36to41', 'MC42to47', 'MC48to53', 'MC54toSA', 'MCSA', 'MFCCBto5', 'MFC
C6to11', 'MFCC12to17', 'MFCC18to23', 'MFCC24to29', 'MFCC30to35', 'MFCC36to41', 'MFCC
42to47', 'MFCC48to53', 'MFCC54toSA', 'MFCCSA', '_75CBto5', '_75C6to11', '_75C12to1
7', '_75C18to23', '_75C24to29', '_75C30to35', '_75C36to41', '_75C42to47', '_75C48to5
3', '_75C54toSA', '_75CSA', '_75FCCBto5', '_75FCC6to11', '_75FCC12to17', '_75FCC18to
23', '_75FCC24to29', '_75FCC30to35', '_75FCC36to41', '_75FCC42to47', '_75FCC48to53',
'_75FCC54toSA', '_75FCCSA', 'iMCBto5', 'iMC6to11', 'iMC12to17', 'iMC18to23', 'iMC24t
o29', 'iMC30to35', 'iMC36to41', 'iMC42to47', 'iMC48to53', 'iMC54toSA', 'iMCSA', 'iMF
CCBto5', 'iMFCC6to11', 'iMFCC12to17', 'iMFCC18to23', 'iMFCC24to29', 'iMFCC30to35',
'iMFCC36to41', 'iMFCC42to47', 'iMFCC48to53', 'iMFCC54toSA', 'iMFCCSA', 'i_75CBto5',
'i_75C6to11', 'i_75C12to17', 'i_75C18to23', 'i_75C24to29', 'i_75C30to35', 'i_75C36to
41', 'i_75C42to47', 'i_75C48to53', 'i_75C54toSA', 'i_75CSA', 'i_75FCCBto5', 'i_75FCC
6to11', 'i_75FCC12to17', 'i_75FCC18to23', 'i_75FCC24to29', 'i_75FCC30to35', 'i_75FCC
36to41', 'i_75FCC42to47', 'i_75FCC48to53', 'i_75FCC54toSA', 'i_75FCCSA', 'MCInfant',
'MCInfant_flag', 'MCToddler', 'MCToddler_flag', 'MCPreschool', 'MCPreschool_flag',
'_75CInfant', '_75CInfant_flag', '_75CToddler', '_75CToddler_flag', '_75CPreschool',
'_75CPreschool_flag', 'MFCCInfant', 'MFCCInfant_flag', 'MFCCToddler', 'MFCCToddler_f
lag', 'MFCCPreschool', 'MFCCPreschool_flag', '_75FCCInfant', '_75FCCInfant_flag', '_
75FCCToddler', '_75FCCToddler_flag', '_75FCCPreschool', '_75FCCPreschool_flag']
```

```
In [12]: #imputed values will be used so remove "imputed or not" binary columns
df = df.loc[:, ~df.columns.str.startswith('i')]
print(list(df.columns))
```

```
[ 'State_Name', 'State_Abbreviation', 'County_Name', 'County_FIPS_Code', 'StudyYear',
'UNR_16', 'FUNR_16', 'MUNR_16', 'UNR_20to64', 'FUNR_20to64', 'MUNR_20to64', 'FLFPR_2
0to64', 'FLFPR_20to64_Under6', 'FLFPR_20to64_6to17', 'FLFPR_20to64_Under6_6to17', 'M
LFPR_20to64', 'PR_F', 'PR_P', 'MHI', 'ME', 'FME', 'MME', 'MHI_2018', 'ME_2018', 'FME
_2018', 'MME_2018', 'TotalPop', 'OneRace', 'OneRace_W', 'OneRace_B', 'OneRace_I', 'O
neRace_A', 'OneRace_H', 'OneRace_Other', 'TwoRaces', 'Hispanic', 'Households', 'H_Un
der6_BothWork', 'H_Under6_FWork', 'H_Under6_MWork', 'H_Under6_SingleM', 'H_6to17_Bot
hWork', 'H_6to17_Fwork', 'H_6to17_Mwork', 'H_6to17_SingleM', 'EMP_M', 'MEMP_M', 'FEM
P_M', 'EMP_Service', 'MEMP_Service', 'FEMP_Service', 'EMP_Sales', 'MEMP_Sales', 'FEM
P_Sales', 'EMP_N', 'MEMP_N', 'FEMP_N', 'EMP_P', 'MEMP_P', 'FEMP_P', 'MCBto5', 'MC6to
11', 'MC12to17', 'MC18to23', 'MC24to29', 'MC30to35', 'MC36to41', 'MC42to47', 'MC48to
53', 'MC54toSA', 'MCSA', 'MFCCBto5', 'MFCC6to11', 'MFCC12to17', 'MFCC18to23', 'MFCC2
4to29', 'MFCC30to35', 'MFCC36to41', 'MFCC42to47', 'MFCC48to53', 'MFCC54toSA', 'MFCCS
A', '_75CBto5', '_75C6to11', '_75C12to17', '_75C18to23', '_75C24to29', '_75C30to35',
'_75C36to41', '_75C42to47', '_75C48to53', '_75C54toSA', '_75CSA', '_75FCCBto5', '_75
FCC6to11', '_75FCC12to17', '_75FCC18to23', '_75FCC24to29', '_75FCC30to35', '_75FCC36
to41', '_75FCC42to47', '_75FCC48to53', '_75FCC54toSA', '_75FCCSA', 'MCInfant', 'MCIn
fant_flag', 'MCToddler', 'MCToddler_flag', 'MCPreschool', 'MCPreschool_flag', '_75CI
nfant', '_75CInfant_flag', '_75CToddler', '_75CToddler_flag', '_75CPreschool', '_75C
Preschool_flag', 'MFCCInfant', 'MFCCInfant_flag', 'MFCCToddler', 'MFCCToddler_flag',
'MFCCPreschool', 'MFCCPreschool_flag', '_75FCCInfant', '_75FCCInfant_flag', '_75FCCT
oddler', '_75FCCToddler_flag', '_75FCCPreschool', '_75FCCPreschool_flag']
```

```
In [14]: print("Dataset shape:", df.shape)
         print("\nData types:\n", df.dtypes.head())
```

Dataset shape: (34567, 128)

Data types:

State_Name	object
State_Abbreviation	object
County_Name	object
County_FIPS_Code	int64
StudyYear	int64

dtype: object

```
In [16]: #number of states, counties, years
         print("Unique states:", df["State_Name"].nunique())
         print("Unique counties:", df["County_Name"].nunique())
         print("Years covered:", df["StudyYear"].unique())
```

Unique states: 51

Unique counties: 1879

Years covered: [2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018]

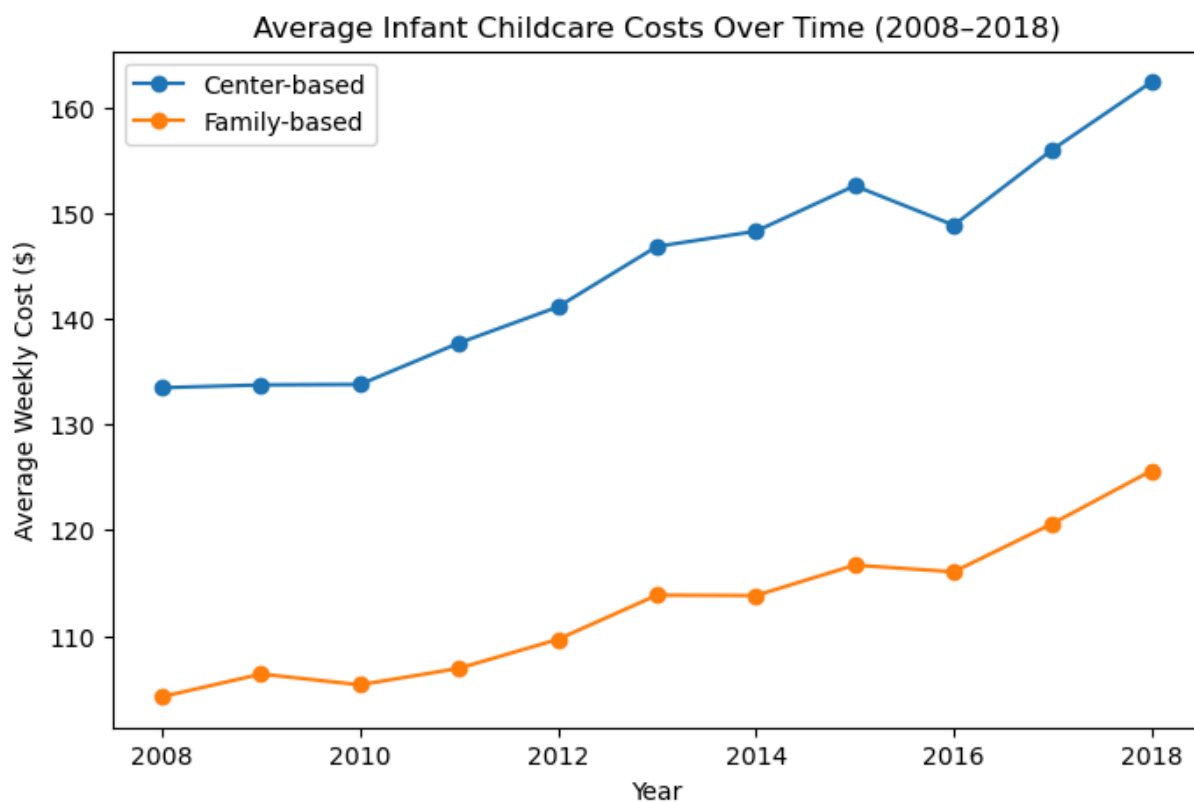
```
In [20]: #null values
         missing_counts = df.isnull().sum()
         missing_sorted = missing_counts.sort_values(ascending=False)
         print(missing_sorted.head())
```

_75FCCPreschool_flag	11184
_75FCCSA	11184
_75FCC48to53	11184
_75FCC42to47	11184
_75FCC36to41	11184

dtype: int64

```
In [22]: #average infant care cost overtime
avg_by_year = df.groupby("StudyYear")[["MCInfant", "MFCCInfant"]].mean()

avg_by_year.plot(marker="o", figsize=(8,5))
plt.title("Average Infant Childcare Costs Over Time (2008-2018)")
plt.xlabel("Year")
plt.ylabel("Average Weekly Cost ($)")
plt.legend(["Center-based", "Family-based"])
plt.show()
```



```
In [24]: #which states are the most or least expensive on average
state_avg = df.groupby("State_Name")[["MCInfant", "MFCCInfant"]].mean().sort_values
print("Top 10 most expensive states:\n", state_avg.head(10))
print("\nTop 10 least expensive states:\n", state_avg.tail(10))
```

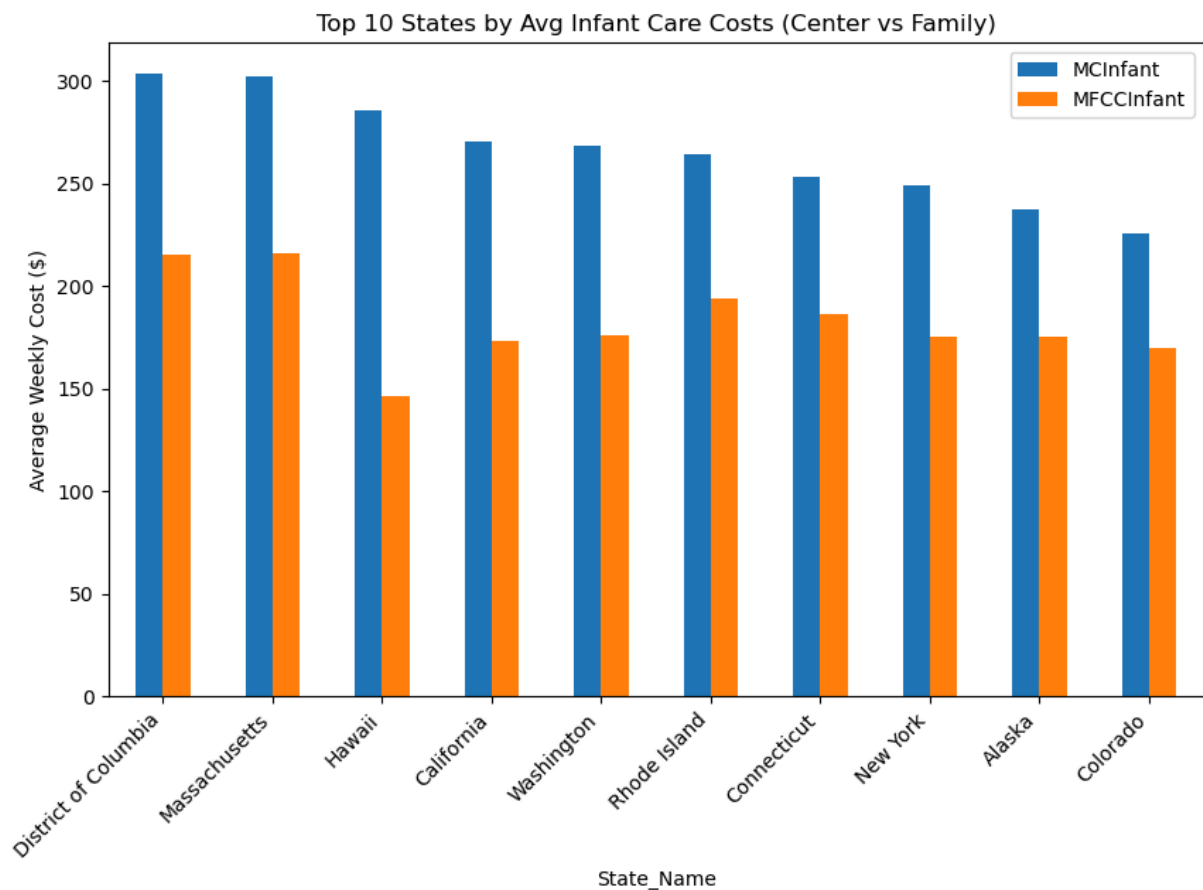
Top 10 most expensive states:

	MCInfant	MFCCInfant
State_Name		
District of Columbia	303.580000	215.770000
Massachusetts	302.033506	215.937662
Hawaii	286.011250	146.754375
California	270.840031	173.336489
Washington	268.560210	176.157832
Rhode Island	264.416000	194.266000
Connecticut	253.238636	186.534091
New York	249.280645	175.319516
Alaska	237.352857	175.299333
Colorado	226.085938	169.735938

Top 10 least expensive states:

	MCInfant	MFCCInfant
State_Name		
Kentucky	112.053523	99.130326
Louisiana	111.714549	90.097691
Georgia	110.534591	94.528302
Alabama	105.752958	91.819281
Arkansas	104.916667	90.583333
Kansas	102.285455	83.715584
South Dakota	96.890480	79.558707
Mississippi	78.620569	66.096626
Indiana	NaN	NaN
New Mexico	NaN	NaN

```
In [26]: state_avg.head(10).plot(kind="bar", figsize=(10,6))
plt.title("Top 10 States by Avg Infant Care Costs (Center vs Family)")
plt.ylabel("Average Weekly Cost ($)")
plt.xticks(rotation=45, ha="right")
plt.show()
```



```
In [30]: #for just 2018
df_2018 = df[df["StudyYear"] == 2018]

state_avg_2018 = (
    df_2018.groupby("State_Name")[["MCInfant", "MFCCInfant"]]
    .mean()
    .sort_values("MCInfant", ascending=False)
)

print("Top 10 most expensive states (2018):\n", state_avg_2018.head(10))
print("\nTop 10 least expensive states (2018):\n", state_avg_2018.tail(10))

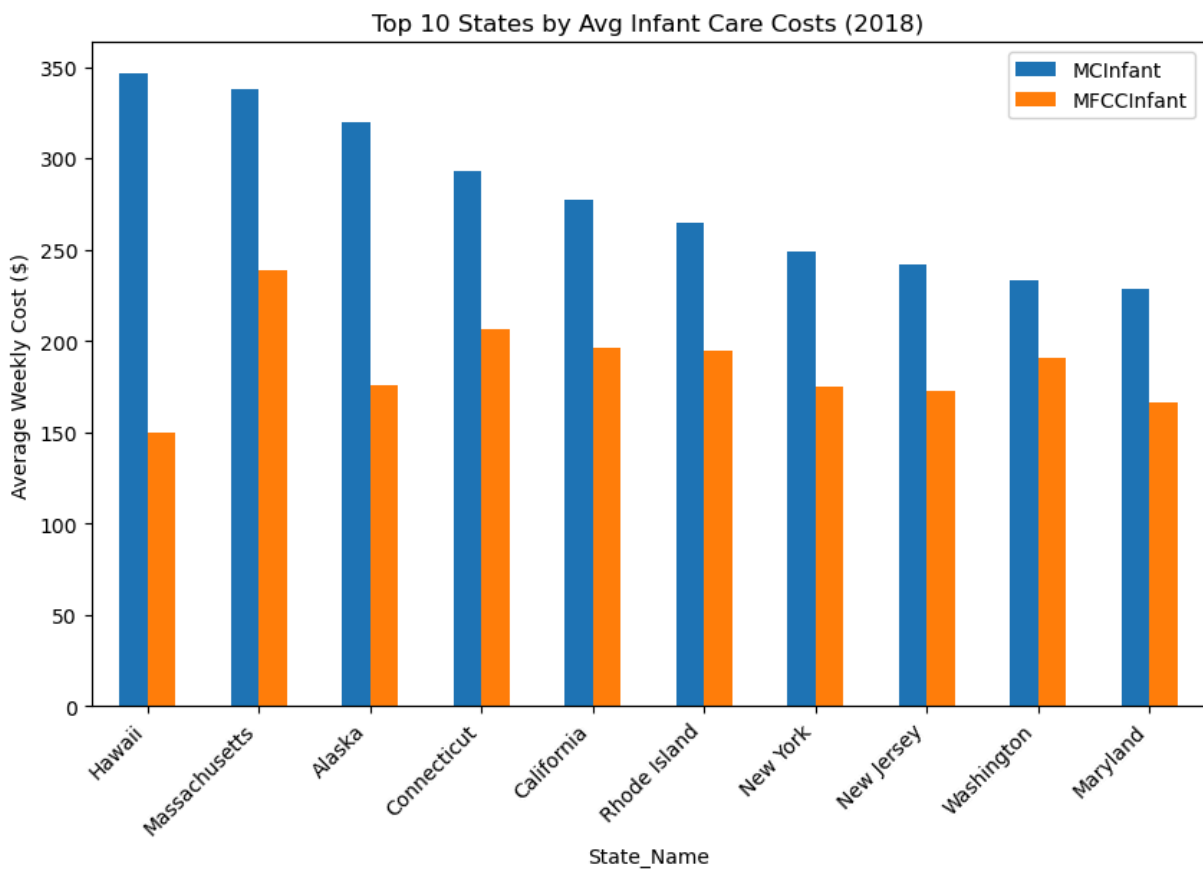
state_avg_2018.head(10).plot(kind="bar", figsize=(10,6))
plt.title("Top 10 States by Avg Infant Care Costs (2018)")
plt.ylabel("Average Weekly Cost ($)")
plt.xticks(rotation=45, ha="right")
plt.show()
```

Top 10 most expensive states (2018):

State_Name	MCInfant	MFCCInfant
Hawaii	346.150000	150.000000
Massachusetts	338.142857	238.571429
Alaska	319.814286	175.960000
Connecticut	293.375000	206.250000
California	277.348793	196.119138
Rhode Island	264.416000	194.266000
New York	249.280645	175.319516
New Jersey	242.212857	172.739524
Washington	233.018974	190.473333
Maryland	228.526667	166.023750

Top 10 least expensive states (2018):

State_Name	MCInfant	MFCCInfant
Colorado	NaN	NaN
District of Columbia	NaN	NaN
Florida	NaN	NaN
Georgia	NaN	NaN
Indiana	NaN	NaN
Iowa	NaN	NaN
Louisiana	NaN	NaN
Montana	NaN	NaN
Nevada	NaN	NaN
New Mexico	NaN	NaN



```
In [32]: #get rid of null values from count
state_avg_2018 = (
```

```

df_2018.groupby("State_Name")[["MCInfant", "MFCCInfant"]]
    .mean()
    .dropna() # remove states with NaN values
    .sort_values("MCInfant", ascending=False)
)

print("\nTop 10 least expensive states (2018):\n", state_avg_2018.tail(10))

```

Top 10 least expensive states (2018):

	MCInfant	MFCCInfant
State_Name		
Missouri	140.429333	110.368000
South Carolina	137.907609	106.195652
West Virginia	133.079818	110.899273
Idaho	129.934091	112.541136
Kentucky	122.887083	100.572917
Alabama	117.653881	98.824925
Kansas	115.835238	97.359048
Arkansas	108.750000	93.750000
South Dakota	108.454848	87.106212
Mississippi	70.064756	60.699146

```

In [36]: #childcare cost vs working mothers
#state/year averages
state_year_avg = (
    df.groupby(["StudyYear", "State_Name"])[["MCInfant", "FLFPR_20to64_Under6"]]
    .mean()
    .reset_index()
)

#all states
national_avg = state_year_avg.groupby("StudyYear")[["MCInfant", "FLFPR_20to64_Under6"]]

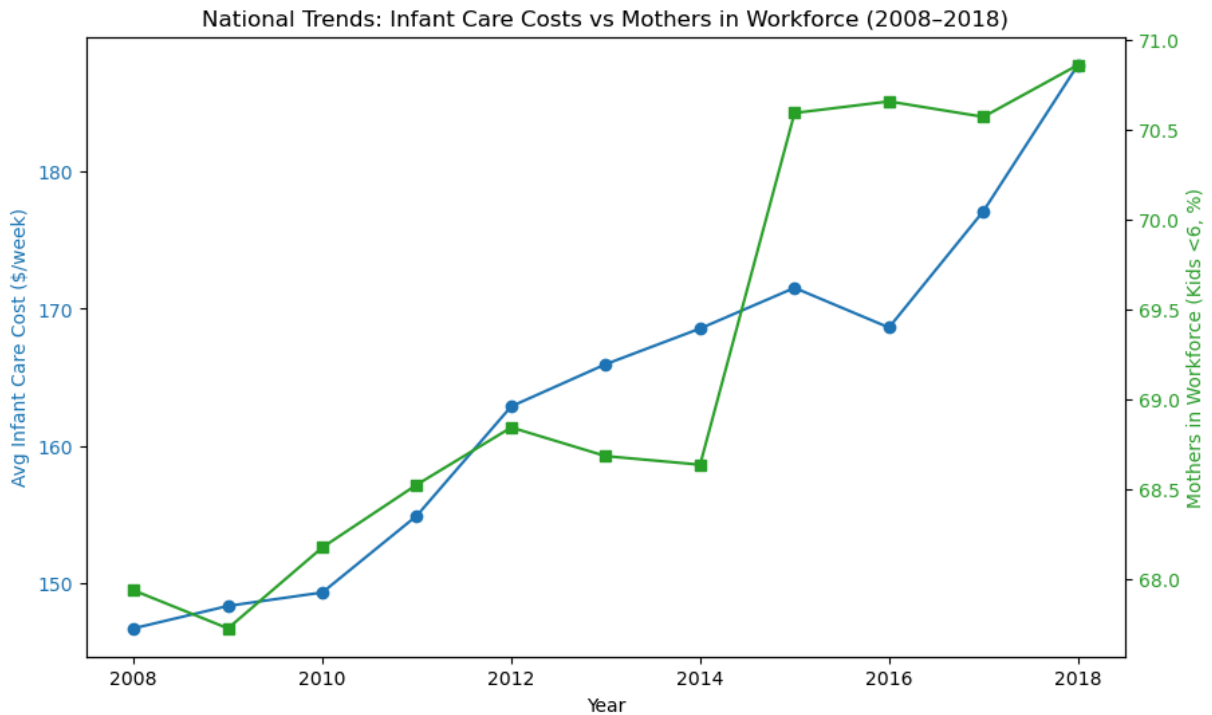
fig, ax1 = plt.subplots(figsize=(10,6))

#childcare cost
ax1.plot(national_avg.index, national_avg["MCInfant"], color="tab:blue", marker="o")
ax1.set_xlabel("Year")
ax1.set_ylabel("Avg Infant Care Cost ($/week)", color="tab:blue")
ax1.tick_params(axis="y", labelcolor="tab:blue")

#mothers in workforce
ax2 = ax1.twinx()
ax2.plot(national_avg.index, national_avg["FLFPR_20to64_Under6"], color="tab:green")
ax2.set_ylabel("Mothers in Workforce (Kids <6, %)", color="tab:green")
ax2.tick_params(axis="y", labelcolor="tab:green")

plt.title("National Trends: Infant Care Costs vs Mothers in Workforce (2008-2018)")
plt.show()

```

```
In [38]: #unemployment vs infant care cost
#national averages by year
national_unemp = df.groupby("StudyYear")[["UNR_20to64", "FUNR_20to64", "MUNR_20to64"]

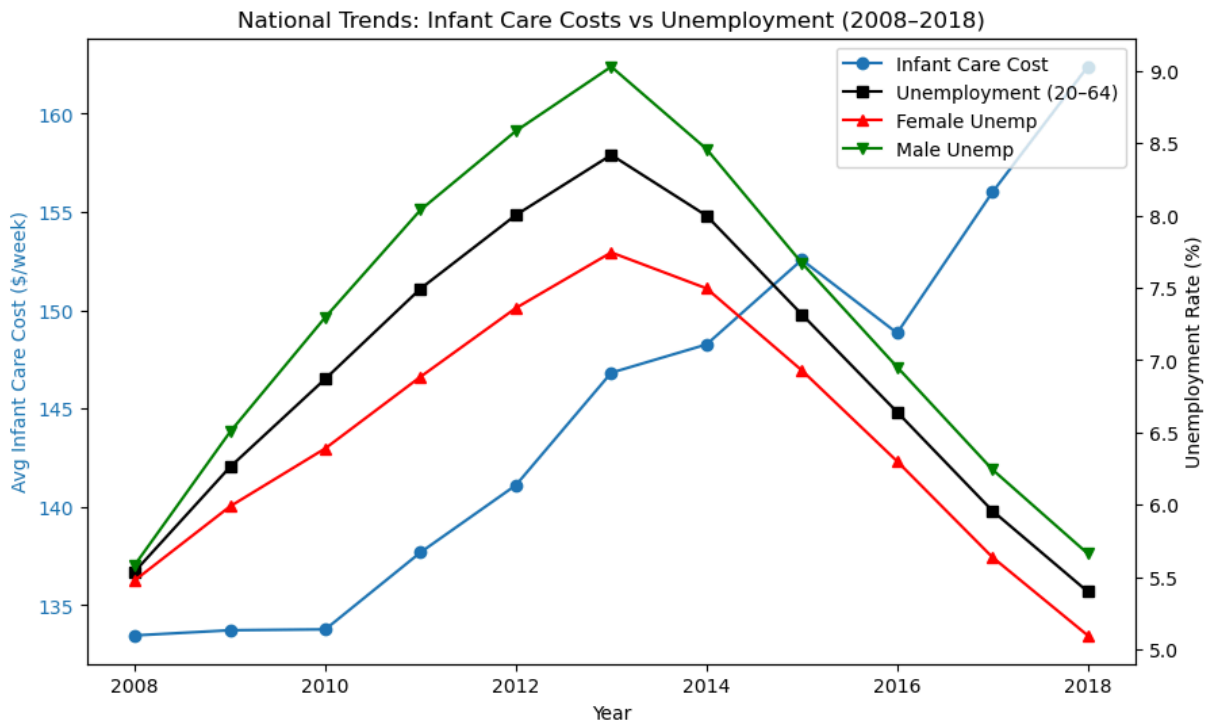
fig, ax1 = plt.subplots(figsize=(10,6))

#infant care cost
ax1.plot(national_unemp.index, national_unemp["MCInfant"], color="tab:blue", marker="o")
ax1.set_xlabel("Year")
ax1.set_ylabel("Avg Infant Care Cost ($/week)", color="tab:blue")
ax1.tick_params(axis="y", labelcolor="tab:blue")

#unemployment rates on second y-axis
ax2 = ax1.twinx()
ax2.plot(national_unemp.index, national_unemp["UNR_20to64"], color="black", marker="o")
ax2.plot(national_unemp.index, national_unemp["FUNR_20to64"], color="red", marker="o")
ax2.plot(national_unemp.index, national_unemp["MUNR_20to64"], color="green", marker="o")
ax2.set_ylabel("Unemployment Rate (%)")
ax2.tick_params(axis="y")

lines, labels = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax2.legend(lines + lines2, labels + labels2, loc="upper right")

plt.title("National Trends: Infant Care Costs vs Unemployment (2008-2018)")
plt.show()
```



```
In [68]: #childcare burden as % of median household income
#annualize childcare costs before dividing by annual income
df["Burden_CenterInfant"] = (df["MCInfant"] * 52 / df["MHI"]) * 100
df["Burden_FCCInfant"] = (df["MFCCInfant"] * 52 / df["MHI"]) * 100
print(df[["MCInfant", "MHI", "Burden_CenterInfant"]].head())
```

	MCInfant	MHI	Burden_CenterInfant
0	104.95	50837.0	10.735095
1	105.11	51463.0	10.620679
2	105.28	53255.0	10.279899
3	105.45	53899.0	10.173473
4	105.61	53773.0	10.212783

```
In [70]: national_burden = df.groupby("StudyYear")[["Burden_CenterInfant", "Burden_FCCInfant"]

ax = national_burden.plot(marker="o", figsize=(10,6))
plt.title("National Trends: Infant Childcare Burden (2008–2018)")
plt.xlabel("Year")
plt.ylabel("Childcare Burden (% of Median Household Income)")
plt.axhline(7, color="red", linestyle="--", label="Affordability Benchmark (7%)")
plt.ylim(0, 25)
plt.legend()
plt.show()
```

